

Roll No. _____

B.Tech IV sem (Main/Back) Exam April-May 2026
Artificial Intelligence & Data Science
4AID4-07 Data Communication and Computer Networks
Paper Code : 4E1307
CS, IT, AID, CAI, CCS, CDS, CIT, CSR

Maximum Marks: 70

Time: 3 Hours

Attempt all ten questions from Part A, five questions out of seven questions from Part B and three questions out of five questions from Part C.

Schematic diagrams must be shown wherever necessary. Any data you feel missing suitably be assumed and stated clearly. Units of quantities used/ calculated must be stated clearly.

Use of following supporting material is permitted during examination. (Mentioned in form No.205)

1. _____

2. _____

Part A (Answer should be given up to 25 words only)**All questions are compulsory**

T-453

- Q.1. Differentiate Analog and Digital signal.
 Q.2. What is peer - to - peer process?
 Q.3. Define piggybacking?
 Q.4. Differentiate between single bit error and burst bit error.
 Q.5. What is Jitter?
 Q.6. What is the use of RJ-45 connector?
 Q.7. Write uses of Leaky Bucket Algorithm.
 Q.8. Write any two differences between UDP and TCP.
 Q.9. Differentiate gateway and routers.
 Q.10. Write use of checksum to find errors in data packets.

10 x 2 = 20**Part B (Analytical/Problem solving questions)****Attempt any Five questions**

- Q.1. Explain OSI reference model with suitable diagram.
 Q.2. What is topology? Explain types of topologies with suitable diagram.
 Q.3. Describe selective repeat ARQ with example. T-453
 Q.4. Explain ARP and RARP address mapping protocols.
 Q.5. Consider the delay of pure ALOHA versus slotted ALOHA at low load. Which one is less?
Explain your answer.
 Q.6. Write short note on
 (a) CSMA/CD
 (b) Architecture of WWW
 Q.7. What is guided transmission media? Explain Coaxial cable.

5x 4 = 20

Part C (Descriptive/Analytical/Problem Solving/Design question)

Attempt any three questions

- Q.1. Explain the working of the following CSMA protocol in detail.
- (a) 1-Persistent
 - (b) Non- Persistent
 - (c) P- Persistent
- Q.2. Explain TCP header format. Differentiate TCP and UDP.
- Q.3. What is ALOHA? Explain ¹⁻⁴⁵⁰ pure and slotted ALOHA. A pure aloha network transmits 200 bits frames on a shared channel of 200 kbps. What is the throughput if the system produces 1000 frames per second?
- Q.4. Write short note on -
- (a) ~~DNS~~
 - (b) Sliding window protocol
 - (c) ~~Stop and wait ARQ~~
- Q.5. Explain the IPv4 header with IPv6 header. Also write the difference between IPv4 and IPv6.
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3x10= 30